PREM KASHYAP CHILAKAMARTHI

prem.kashyap.0206@gmail.com | +91 9502645945 | prem-kashyap.pages.dev | prem-kashyap-chilakamarthi | github.com/premkashyap004

Professional Summary

Generative AI Developer with 3 years of production experience building multi-agent systems, RAG pipelines, and real-time AI applications. Reduced incident resolution time by 40–50% and content creation effort by 30% through LangGraph orchestration and GPU-based LLM inference (vLLM, TensorRT-LLM) deployed on on-premises and cloud infrastructure at scale.

Technical Skills

LLM Frameworks	LangChain, LlamaIndex, LangGraph, LangFuse, HuggingFace, OpenAI
LLM Inference & Serving	vLLM, TensorRT-LLM, XInference, Ollama
AI Concepts	RAG, Multi-Agent Systems, Autonomous Agents, Prompt Engineering, MCP, Skills
Programming Languages	Python, JavaScript, TypeScript
Backend	FastAPI, Flask, Node.js, Socket.IO, FastRTC
Frontend	React, HTML, CSS
Cloud & DevOps	AWS, GCP, Docker, Kubernetes, GitHub Actions
Databases & Vectors	PostgreSQL, MongoDB, Redis, Chroma, Pinecone, FAISS

Work Experience

Generative AI Developer, **Sify Technologies** July 2023 – Present

- Developed production RAG and multi-agent solutions (LangChain, LlamaIndex, LangGraph) with full-stack AI applications (FastAPI, React). Integrated LangFuse observability to streamline monitoring, prompt tuning, and debugging workflows.
- Deployed and managed LLM serving infrastructure using vLLM and TensorRT-LLM, and maintained multiple AI applications across AWS, GCP, and on-premises servers.

Projects

AI-Driven IT Operations Automation – Python, FastAPI, MongoDB 2026

- Reduced manual Linux operations effort significantly by developing AI-driven automation workflows using an in-house orchestration platform for access control, time-based provisioning, permissions management, and system monitoring across production environments, leveraging SSH-based Linux command execution and SAP RFC integrations.
- Ensured security and compliance by integrating policy-driven decision-making, audit logging, and human-in-the-loop approval workflows across all automated processes.

Ticket Resolution Automation System – LangGraph, Python, MCP, FastAPI, Redis, React 2025

- Reduced manual incident resolution effort by 60% and accelerated resolution times by 40–50% by building a multi-agent LangGraph system to automate cloud incident triage and OS-level remediation across environments.
- Improved operational safety and auditability by implementing human-in-the-loop approval workflows and an MCP-based chatbot for real-time, tool-driven operational insights and also integrated with Langfuse for observability and traceability.

Scout AI – Companion Chatbot for Children – GPT-4o, Python, FastRTC, React 2025

- Reduced child onboarding time and increased engagement by building a real-time AI companion (ages 4–12) using GPT-4o and FastRTC, delivering low-latency bedtime stories, homework help, and adaptive storytelling.
- Improved learning retention and personalisation by architecting a multi-agent context-switching system with dynamic roleplay, AI-generated quizzes, and age-appropriate content guardrails for safe, immersive interactions.

Skillflo – GenAI Learning Platform – LangChain, RAG, Chroma, Flask, Node.js, AWS 2024

- Cut content creation effort by over 30% by architecting a microservice GenAI applications using LangChain and LlamaIndex on AWS, enabling automated text-to-video generation, media assets management, storyboard creation, and RAG-based assessment building.
- Implemented multi-tenant ChromaDB-based retrieval pipelines for RAG applications, supporting cross-domain semantic search across storyboards and media assets while improving AI chatbot accuracy through source grounding and fact-checking.

Education

B.E. Computer Science and Engineering Aug 2019 – Apr 2023
Sri Sairam Engineering College, Chennai CGPA: 9.28/10.0

Certifications

AWS Certified AI Practitioner – View certificate May 2026 – May 2029
AWS Certified Developer Associate – View certificate Apr 2023 – Apr 2026
Oracle Generative AI Certificate – View certificate Jul 2024 – Jul 2026